

AI PROJECT

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Semester	6th
Project	AI Sentiment Analysis

Artificial Intelligence

1. Introduction to AI Sentiment Analysis

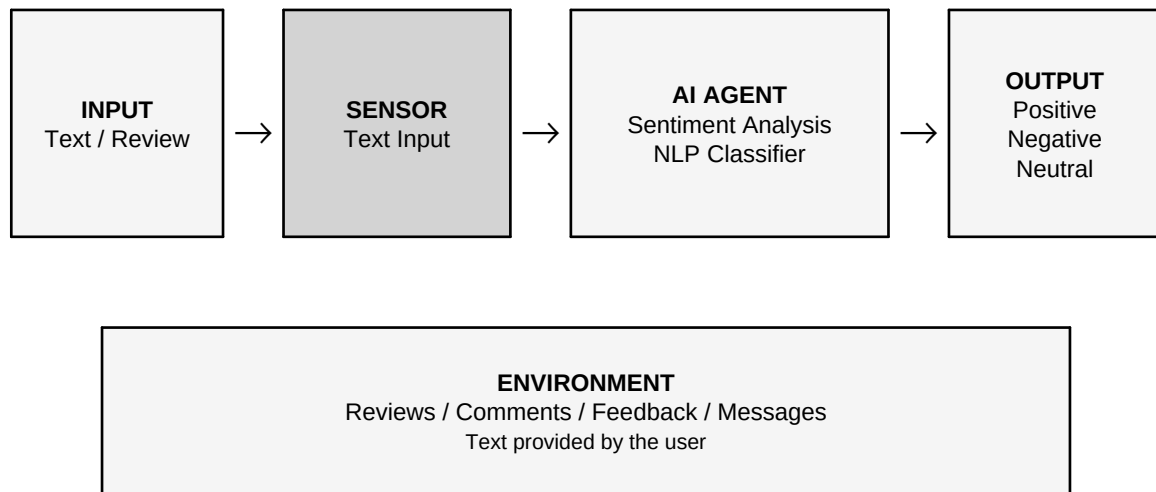
Sentiment Analysis is an Artificial Intelligence and Natural Language Processing (NLP) technique used to identify the emotional opinion expressed in a piece of text. The system analyzes text such as a review, comment, feedback or message and classifies its sentiment.

A basic Sentiment Analysis Agent can classify text into three common categories:

Sentiment	Meaning	Example
Positive	The text expresses a favorable opinion.	"The product is excellent."
Negative	The text expresses an unfavorable opinion.	"The service was terrible."
Neutral	The text is factual or does not show a clear opinion.	"The package arrived today."

Project Goal: The goal of the AI Sentiment Analysis Agent is to read user-provided text, analyze its language features and automatically determine whether the sentiment is positive, negative or neutral.

2. Graphical Representation of the Agent



Flow: Text/Review → Sensor → AI Sentiment Analysis Agent → Classification → Positive / Negative / Neutral.

The environment provides text to the agent. The sensor receives the text, the AI agent analyzes the text using NLP/classification, and the actuator produces the final sentiment label.

3. Agent Formulation

Agent formulation describes what information the agent receives, where it operates, how it senses the environment, what action it performs, and what goal it tries to achieve.

Component	Description
Environment	Online reviews, social-media comments, customer feedback, messages or other text-based environments.
Sensors	Text input from the user; the agent receives words, sentences and relevant language features.
Actuators	Displays or assigns the predicted sentiment label: Positive, Negative or Neutral.
Percepts	Words, phrases, sentence structure, opinion-related terms and other text features observed from the input.
Goal	Accurately identify the sentiment expressed in the input text.
Actions	Analyze the text, classify its sentiment and return the predicted category.

Agent Type / Environment Properties

Partially Observable: The agent sees only the text supplied to it and may not know the complete context or intention of the writer.

Stochastic: Language can be ambiguous, and the same word may have different meanings in different contexts.

Sequential: A system may process a sequence of reviews or comments, although each text can also be classified independently.

Dynamic: New reviews, comments and messages can arrive while the system is operating.

Discrete: The output categories are discrete labels: Positive, Negative and Neutral.

Known: When a trained dataset/model is available, the agent uses learned patterns to classify new text.

4. PEAS Framework Evaluation

PEAS stands for Performance Measure, Environment, Actuators and Sensors. It is used to describe the task environment and expected behavior of an intelligent agent.

PEAS Element	AI Sentiment Analysis Agent
P — Performance Measure	High classification accuracy, correct identification of positive/negative/neutral sentiment, low misclassification rate, fast response and reliable results.
E — Environment	Online product reviews, customer feedback, social-media comments, messages and other text-based platforms.
A — Actuators	Display the predicted sentiment, label the text as Positive/Negative/Neutral, and optionally show a sentiment score or confidence.
S — Sensors	Receive text input and capture words, phrases and other useful language features from the text.

Example Evaluation

Input Text	Predicted Sentiment
"I really enjoyed this product."	Positive
"The service was disappointing."	Negative
"The order arrived on Monday."	Neutral

Conclusion

The AI Sentiment Analysis Agent is a simple practical AI agent that uses text as its input, analyzes the expressed opinion and produces a sentiment classification. Its environment, sensors and actuators can be clearly defined, and its task can be evaluated using the PEAS framework.